

ST 3007 – Operational Research

Tutorial 2

Binary Integer Programming

1. Suppose that you have 7 full wine bottles, 7 half-full, and 7 empty. You would like to divide the 21 bottles among three individuals so that each will receive exactly 7. Additionally, each individual must receive the same quantity of wine. Express the problem as an ILP constraint and find a solution. (Hint: Use a dummy objective function with all zero coefficients.)
2. An eccentric sheikh left a will to distribute a herd of camels among his three children: Tarek receives at least one-half of the herd, Sharif gets at least one third, and Maisa gets at least one-seventh. The remainder goes to charity. The will does not specify the size of the herd except to say that it is an odd number of camels and that the named charity receives exactly one camel. Use ILP to determine how many camels the sheikh left in the estate and how many each child got.
3. BC is a trucking company that delivers loads on a daily basis to five customers. The following list provides the customers associated with each route:

Route	Customers served on the route
1	3,2
2	5,3,4
3	2,5,1,3
4	2,3,5
5	1,4,2
6	1,3,5

The segments of each route are dictated by the capacity of the truck delivering the loads. For example, on route 1, the capacity of the truck is sufficient to deliver the loads to customers 3 and 2 only. The following table lists distances (in miles) between the truck terminal (ABC) and the customers.

	ABC	1	2	3	4	5
ABC	0	10	12	16	9	8
1		0	32	8	17	10
2			0	14	21	20
3				0	15	18
4					0	11
5						0

The objective is to determine the least distance needed to make the daily deliveries to all five customers. Though the solution may result in a customer being served by more than one route,

an approximation in the implementation phase assumes that only one such route is used. Formulate the problem as an ILP and find the optimum solution.

4. A company is attempting to decide the mix of products which it should produce during the next week. It has seven products, each with a profit per unit (\$) and a production time (man-hours) per unit as shown in Table 1.

Table 1: Profits and production time of each product

Product	Profit (\$ per unit)	production time (man-hours per unit)
1	10	1
2	22	2
3	35	3.7
4	19	2.4
5	55	4.5
6	10	0.7
7	115	9.5

The company has the following additional restrictions:

- The company has 720 man-hours available during next week.
- If product 7 is produced, an additional fixed cost of \$2000 is incurred (Hint: use a binary integer variable).
- Each unit of product 2 that is produced over 100 units requires a production time of 3.0 man-hours instead of 2.0 man-hours. That is, producing 101 units of product 2 requires $100(2.0) + 1(3.0)$ man-hours (Hint: define a variable to indicate how many units of product 2 produced more than 100 units).
- Weekly demand for products 1, 2, 3 and 4 depends on a certain market condition. If that condition is satisfied, the total combined weekly demand of products 1 and 3 is 80 units (no effect on products 2 and 4 here). If that condition is not satisfied, the total combined weekly demand of products 2 and 4 is 85 units (no effect on products 1 and 3 here).

Clearly stating all the decision variables, formulate this scenario as an Integer Linear Programming Problem (ILPP) to decide how many units of each product are to be produced during the next week. You **do not** have to solve the problem.

(Exam Question – 2022)